

SUMMARY

Computer Engineering Graduate with hands-on experience in Legged locomotion, Reinforcement learning, Embedded systems, State estimation, Computer Vision, Path Planning, ROS, Test and deployment.

EDUCATION

California State University, Fullerton, MS in Computer Engineering

Aug 2023 - May 2025

Relevant Subjects- Artificial Intelligence, RTOS for embedded systems, Artificial Neural Networks, Robot manipulation and planning. **Graduate Thesis** - Legged Robotics under Dr Jidong Huang

SKILLS

Linux, C++, Python, ROS, Git, Docker, Gazebo, Pybullet, Tensorflow, OpenCV, Pandas, PyTorch, Scikit-learn, FD CAN, I2C, 3D Printing, Prototyping

WORK EXPERIENCE

Graduate Research Assistant

(Sep 2024 – May 2025)

Institute of Navigation (ION) Lab, Advisor – Dr. Yu Bai, Dr. Doina Bein

- Built an open source low-cost (under \$1000) 12 DOF 3D-printed **quadruped robot** capable of walking on uneven terrain
- Integrated **FSRs and IMU** to feed sensory data to linear policy and successfully tested on uneven ground
- Mentored 100+ underrepresented engineering and computer science undergraduates in C++, Python, machine learning and Robotics using Raspbot 4 wheeled robot kits at ACCESS workshops

NASA L'SPACE PROGRAM, Arizona State University

(Mar 2024- Aug 2024)

- Led the design and proposal of an autonomous docking interface between a Mars rover(UGV)and aerial drone(UAV), aimed at enabling power transfer and mitigating energy limitations with the UAV guiding path planning to optimize rover traversal

Robotics Software Engineer

(Feb 2023 – June 2023)

ARTPARK, Indian Institute of Science, Bangalore

- Simulated and trained quadruped walking policies using the Motion Imitation framework in PyBullet
- Tested a 1:16 multi-stage planetary gear reduction delivering torque output of 30+Nm.

Co-founder Blufos

(Aug 2020 – Dec 2022)

- Created Python scripts and LabVIEW applications for data logging, sensor calibration, and test automation; implemented serial communication and real-time data visualization.
- Integrated OctoPrint API for remote 3D printer management and job scheduling, improving utilization

PROJECTS

MINIMAL ROBOT DOG - Graduate Thesis

- Researched the effect of emulating **contact sensing and ground reaction forces** on the efficacy of linear control policies using **Augmented Random Search** agent in Pybullet over uneven randomized terrain
- Implemented **sim-to-real** transfer using ROS, leveraging force-sensitive resistors to validate the performance of a trained contact sensing policy in real-world scenarios

SCHVAAN ROBOT DOG

- Built a 12 DOF robot dog with self-built 3D printed **cycloidal actuators of 1:10 gear reduction**
- Tested the accuracy and reliability of actuators for torque, back-drivability and low backlash

HUMAN ROBOT INTERACTION USING POINTING GESTURES

- Developed a real-time system using **OpenCV and PyTorch** that combines **MediaPipe Pose Estimation** with **YOLOv4** object detection to identify objects pointed/gestured at by the user
- Implemented geometric calculations to **estimate ground intersection points** based on body landmarks for enhanced spatial interaction

PUBLICATIONS

- Publication: (ICCAR 2025- Accepted): [Ground contact and reaction force sensing for linear policy control of quadruped robot](#)
- An article about the Schvaan robot was published in a leading National newspaper. – [Link](#)